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Interference-Based Quantum Classification Using Linear State Overlap

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Abstract—Quantum machine learning (QML) classifiers proposed for near-term quantum hardware typically rely on variational quantum circuits (VQCs) or fidelity-based similarity estimation. While these approaches offer high expressivity, they are often hindered by extreme measurement costs, unstable optimization landscapes (e.g., barren plateaus), and significant sensitivity to hardware noise, which collectively limit their practical utility on noisy intermediate-scale quantum (NISQ) devices. This work presents a measurement-efficient, interference-based quantum classification framework that moves away from parameterized circuit training and probability-based decision rules. The proposed approach performs inference using a fixed (non-trainable) quantum interference circuit, where the classification decision is derived directly from the real part of the linear overlap between quantum state representations—a quantity accessed through a physical interference signal. By decoupling learning from quantum execution, we enable classical computation to handle the construction of static class-representative quantum states in Hilbert space. We provide a rigorous mathematical formalization of this linear interference decision primitive, investigate its geometric properties as a hyperplane classifier in Hilbert space, and analyze its shot complexity, showing it requires significantly fewer measurements than fidelity-based methods. Through extensive simulation under realistic noise parameters, we demonstrate that the proposed classifier achieves stable performance. Inference reduces to a sign test on a single ancilla observable, yielding constant-shot decision complexity with respect to system size for fixed confidence and non-vanishing margin. This work establishes linear quantum interference as a robust and efficient decision primitive for the next generation of near-term quantum classification systems.

Index Terms—Quantum Machine Learning, Quantum Classification, Quantum Interference, Linear Overlap, NISQ Devices, Hybrid Quantum-Classical Systems

I. INTRODUCTION

The rapid advancement of quantum hardware has spurred intense interest in quantum machine learning (QML), a field that seeks to leverage quantum mechanical phenomena—such as superposition, entanglement, and interference—to enhance data processing and pattern recognition tasks [1], [2]. Among the various branches of QML, supervised classification has emerged as a primary focus, with researchers proposing diverse architectures including variational quantum classifiers (VQCs), quantum kernel methods, and fidelity-based similarity models [3], [5], [8]. These methods are largely tailored for the Noisy Intermediate-Scale Quantum (NISQ) era, where limited

qubit counts and high gate error rates necessitate hardware-efficient designs.

Despite the proliferation of these models, several fundamental bottlenecks impede their transition from theoretical constructs to practical industrial tools. Variational approaches, for instance, rely on an iterative feedback loop between a quantum processor and a classical optimizer. In this “hybrid” loop, the quantum device evaluates a cost function, and the classical optimizer adjusts the parameters of a quantum circuit accordingly. However, the optimization landscape of VQCs is frequently plagued by barren plateaus—regions where gradients vanish exponentially with the number of qubits—making the training process computationally expensive and often unreliable [4].

Alternative approaches, such as quantum kernel and fidelity-based classifiers, attempt to mitigate the training burden by moving optimization to the classical domain. These models use the quantum device solely to estimate similarities (typically fidelities) between data points. While avoiding quantum gradients, these methods introduce a different challenge: measurement explosion. Estimating the fidelity $|\langle\phi|\psi\rangle|^2$ between two states requires a significant number of measurement shots to achieve the precision necessary for accurate classification, particularly when the states are highly overlapping or nearly orthogonal. Furthermore, fidelity is a quadratic quantity that discards the relative phase between state amplitudes, potentially losing valuable information encoded in the signs of the quantum state coefficients.

This work proposes a paradigm shift in near-term quantum classification. Rather than optimizing parameterized circuits or estimating quadratic probabilities, we ground the classification decision in a linear interference primitive: the real part of the quantum state overlap, $\text{Re}\langle\phi|\psi\rangle$. The proposed framework, termed the Interference-Based Quantum Classifier (IQC), employs a fixed, non-trainable quantum interference circuit for inference. The learning phase—constructing the representative “memory” states for each class—is performed entirely through classical preprocessing of the training data. This strict decoupling of learning and execution ensures that the quantum hardware is used only for the task it is uniquely suited for: performing high-dimensional interference in Hilbert space.

Crucially, the IQC framework offers a path toward constant measurement-shot complexity. Because the decision rule depends only on the sign of the interference signal rather than its precise magnitude, the number of measurements required to classify a sample remains nearly constant, regardless of the precision required for the underlying overlap value. This makes the system inherently more robust to the shot noise and decoherence characteristic of current quantum hardware.

The primary contributions of this work are summarized as follows:

- We introduce **linear quantum interference** as a decision primitive for classification, demonstrating its conceptual advantages over quadratic observables like fidelity.
- We formalize the **Decision Geometry** of the IQC, showing that it naturally implements a hyperplane classifier within the exponentially large Hilbert space of the quantum system.
- We describe a hardware-efficient **Interference Circuit** based on the Hadamard test that directly extracts the required decision signal with minimal gate overhead.
- We provide a comprehensive **Shot-Complexity Analysis**, proving that accurate inference can be achieved with $O(1)$ measurement shots for the majority of classification tasks.
- We conduct **Simulated Experiments** under various noise regimes, validating the robustness and efficiency of the proposed method against standard VQC and kernel baselines.

Section II reviews related work in QML. Section III defines the problem setup and notation. Section IV presents the mathematical formulation of linear interference as a decision primitive. Section V describes the quantum circuit and its measurement properties. Section VI outlines the static learning framework. Section VII presents experimental results, followed by a discussion of limitations in Section VIII and conclusion in Section IX.

II. RELATED WORK

The landscape of quantum machine learning (QML) has evolved rapidly, moving from early theoretical subroutines for linear algebra to hardware-efficient architectures designed for NISQ hardware. This section categorizes the most relevant approaches and highlights the research gaps addressed by the linear interference-based framework.

A. Variational Quantum Classifiers (VQCs)

Variational quantum classifiers represent the most prominent family of NISQ-era machine learning models [3], [8]. VQCs use a parameterized quantum circuit (PQC) as an ansatz, whose parameters are optimized to minimize a classical cost function. While highly flexible, VQCs are notoriously difficult to train. The existence of barren plateaus—regions in the parameter space where gradients vanish exponentially with the number of qubits—often renders gradient-based optimization infeasible for larger systems [4]. Furthermore, VQCs require a high volume of measurement shots at each optimization step

to achieve low-variance gradient estimates, placing a heavy burden on the quantum processor's duty cycle.

B. Quantum Kernel and Similarity-Based Methods

Quantum kernel methods avoid the optimization difficulties of VQCs by mapping data into high-dimensional Hilbert space and computing similarities between pairs of data points [5], [9]. These similarities, typically defined as the fidelity $F(\rho, \sigma) = \text{Tr}(\sqrt{\sqrt{\rho}\sigma\sqrt{\rho}})^2$, are used as kernel entries in classical algorithms like Support Vector Machines (SVMs). While mathematically elegant, kernel methods suffer from quadratic scaling in terms of quantum circuit executions: to classify a new point against a dataset of size N , one must perform $O(N)$ similarity measurements. Moreover, fidelity is a quadratic observable that removes relative phase information, which can be detrimental in cases where the decision boundary depends on the underlying sign structure of the state amplitudes.

C. Quantum State Discrimination and Metric Learning

Quantum state discrimination (QSD) approaches treat classification as the task of identifying which of several known quantum states a given system is in [10], [11]. This perspective shifts the focus from circuit parameters to the geometric arrangement of states in Hilbert space. Relatedly, quantum metric learning attempts to learn a feature map that maximizes the distance between states representing different classes while minimizing it for states of the same class. Our work builds on these geometric foundations but introduces a distinct inference mechanism: we use a fixed interference circuit to extract a linear overlap signal, providing a simpler and more measurement-efficient alternative to full state tomography or fidelity estimation.

D. Interference-Based QML

The use of interference for pattern recognition was pioneered in works such as the distance-based classifier by Schuld et al. [7], [12], which utilized the Hadamard test to compute distances between vectors. However, many early implementations required the use of QRAM or complex state-preparation routines that are difficult to realize on NISQ devices. In contrast, our framework focuses on purely linear interference as a decision primitive using static, pre-computed class centroids, significantly reducing both the circuit depth and the sampling overhead required for practical inference.

III. DATA MAPPING AND QUANTUM FEATURE SPACES

A critical component of any quantum classifier is the mapping of classical data into quantum states. The choice of feature map directly determines the structure of the data in Hilbert space and, consequently, the effectiveness of the linear decision boundary.

TABLE I
COMPARISON OF QUANTUM CLASSIFICATION PARADIGMS

Feature	Variational (VQC)	Kernel (QSVM)	Fidelity-Based	Linear Interference (IQC)
Decision Basis	Probabilities $P(y x)$	High-dim Kernels	Quadratic Overlap F	Linear Overlap $\text{Re}\langle\phi \psi\rangle$
Learning Mode	In-circuit Optimization	Classical SVM	Static/Adaptive	Static Classical Aggregation
Quantum Circuit	Parameterized (Trainable)	Fixed (Feature Map)	Fixed (Swap Test)	Fixed (Hadamard Test)
Shot Complexity	High ($O(M \cdot N)$)	Moderate ($O(N)$)	Moderate ($O(1/\epsilon^2)$)	Low ($O(1)$ for sign)
Phase Preservation	No	No	No	Yes
Optimization Risk	Barren Plateaus	None (in quantum)	None	None

A. Encoding Strategies for Linear Interference

We consider three primary encoding strategies adapted for the IQC framework:

- 1) **Amplitude Encoding:** Maps a normalized vector $\mathbf{x} \in \mathbb{R}^d$ to the amplitudes of a 2^n -dimensional quantum state. This provides the highest information density ($O(2^n)$ features) but requires complex state-preparation circuits. In IQC, amplitude encoding ensures that the linear overlap in Hilbert space is directly proportional to the Euclidean dot product in the input space.
- 2) **Angle Encoding:** Maps each feature x_j to the rotation angle of a qubit, $|\psi\rangle = \bigotimes_j (\cos x_j |0\rangle + \sin x_j |1\rangle)$. This is hardware-efficient but introduces non-linearities (trigonometric functions) into the decision signal.
- 3) **Basis Encoding:** Maps classical bitstrings directly to computational basis states. While simple, it does not exploit the superposition-based interference that defines the IQC's advantage.

B. Linear Separability in High-Dimensional Feature Spaces

According to Cover's Theorem, a complex pattern-classification problem cast in a high-dimensional space non-linearly is more likely to be linearly separable than in a low-dimensional space. By mapping data into the 2^n -dimensional Hilbert space, the IQC leverages this high dimensionality to separate classes using a simple hyperplane.

However, unlike Kernel methods that use an implicit feature map, the IQC works with an **explicit feature map** $\Phi(\mathbf{x})$. The linear overlap $\text{Re}\langle\phi|\psi\rangle$ is computed directly in the state space. The success of the static centroid construction (Section VI) depends on whether the chosen Φ places the class distributions in convex regions that can be separated by a single hyperplane.

IV. LINEAR INTERFERENCE AS A DECISION PRIMITIVE

The core theoretical contribution of this work is the use of linear interference as the primary signal for quantum classification. This section formalizes the distinction between linear and quadratic similarity measures and analyzes the resulting decision geometry in Hilbert space.

A. Linear vs. Quadratic Similarity

In traditional quantum classification frameworks, similarity between states is often defined via the fidelity $F(\phi, \psi) = |\langle\phi|\psi\rangle|^2$. While fidelity is a natural metric for state closeness, its quadratic nature introduces several practical disadvantages for inference on NISQ devices:

- 1) **Phase Erasure:** The absolute square operation $|\cdot|^2$ removes the relative phase and sign information between state amplitudes. In many machine learning contexts, the sign of the overlap (interference) carries critical directional information that is lost in fidelity estimation.
- 2) **Non-linear Decision Boundaries:** In the space of state amplitudes, fidelity induced decision boundaries are second-order surfaces (ellipsoids or hyperboloids). This can lead to overfitting or complex optimization landscapes when trying to separate classes.
- 3) **Shot Noise Sensitivity:** Because fidelity is a probability-like quantity, estimating it to high precision requires a number of measurement shots that scales inversely with the square of the desired error, $O(1/\epsilon^2)$.

In contrast, the linear overlap $\langle\phi|\psi\rangle$ is a complex-valued quantity. In this paper, we focus on its real part:

$$s(\psi) = \text{Re}\langle\phi|\psi\rangle. \quad (1)$$

This quantity preserves the interference pattern between states and, as we show below, enables a much more measurement-efficient decision rule.

B. Decision Function and Labeling Rule

We define the binary classification decision based on the sign of the real overlap between the input state $|\psi\rangle$ and a pre-determined class-representative state $|\phi\rangle$. The predicted label $\hat{y}$ is given by:

$$\hat{y} = \text{sign}(\text{Re}\langle\phi|\psi\rangle). \quad (2)$$

Setting the threshold at zero corresponds to a balanced classification. If prior class probabilities are known, a bias term $b \in \mathbb{R}$ can be added classically:

$$\hat{y} = \begin{cases} +1 & \text{Re}\langle\phi|\psi\rangle + b \geq 0 \\ -1 & \text{Re}\langle\phi|\psi\rangle + b < 0 \end{cases} \quad (3)$$

C. Decision Geometry: The Hilbert Space Hyperplane

Geometrically, the decision rule $\text{Re}\langle\phi|\psi\rangle = 0$ defines a **linear hyperplane** that partitions the 2^n -dimensional Hilbert space into two half-spaces.

To see this, let $|\psi\rangle$ and $|\phi\rangle$ be represented by their amplitude vectors $\mathbf{c}_\psi, \mathbf{c}_\phi \in \mathbb{C}^{2^n}$. The real part of the inner product can be written as:

$$\text{Re}\langle\phi|\psi\rangle = \frac{1}{2} (\langle\phi|\psi\rangle + \langle\psi|\phi\rangle) = \text{Re}(\mathbf{c}_\phi^\dagger \mathbf{c}_\psi). \quad (4)$$

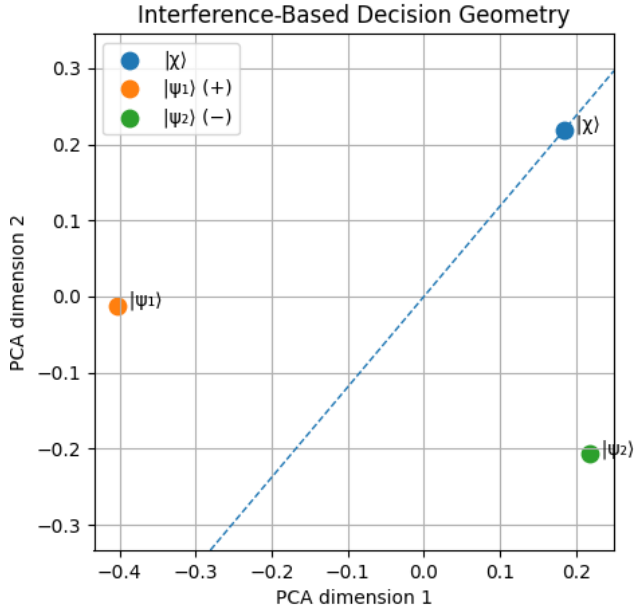


Fig. 1. Comparison of decision boundaries: Linear Interference (left) vs. Fidelity-based Quadratic (right). The interference rule $\text{Re}\langle\phi|\psi\rangle = 0$ induces a linear hyperplane, preserving phase information, while fidelity-based rules result in curved boundaries that discard the state's sign structure.

In the real-valued embedding of Hilbert space (treating $\mathbb{C}^{2^n}$ as $\mathbb{R}^{2 \cdot 2^n}$), this corresponds to the standard dot product between two vectors. Thus, our classifier is effectively a **Quantum Linear Support Vector Machine** or **Centroid Classifier** operating in the feature space defined by the quantum mapping, but with the unique advantage of using a fixed interference circuit for computation.

D. Formal Derivation of the Decision Score

In this subsection, we rigorously derive the relationship between the quantum interference signal and the classification decision. Consider the normalized state $|\psi\rangle = \sum_i a_i |i\rangle$ and the representative state $|\phi\rangle = \sum_i b_i |i\rangle$. The interference term $\text{Re}\langle\phi|\psi\rangle$ can be expanded in terms of the real and imaginary parts of the coefficients:

$$\langle\phi|\psi\rangle = \sum_i b_i^* a_i = \sum_i (\text{Re } b_i - i \text{Im } b_i)(\text{Re } a_i + i \text{Im } a_i) \quad (5)$$

Taking the real part, we obtain:

$$\text{Re}\langle\phi|\psi\rangle = \sum_i (\text{Re } a_i \text{Re } b_i + \text{Im } a_i \text{Im } b_i) \quad (6)$$

This summation is identical to the Euclidean inner product in $\mathbb{R}^{2^{n+1}}$. The decision boundary $\text{Re}\langle\phi|\psi\rangle = 0$ is therefore the locus of all states in $\mathcal{H}$ that are "mutually orthogonal" under this real-part inner product.

V. QUANTUM INTERFERENCE CIRCUIT FOR INFERENCE

The accessibility of the linear overlap signal $\text{Re}\langle\phi|\psi\rangle$ is the defining hardware requirement of our framework. Since standard projective measurements $\langle\psi|O|\psi\rangle$ are quadratic in the state amplitudes, we utilize an ancilla-assisted interference circuit. In this framework, one inference corresponds to executing a single fixed interference circuit on an input state and assigning a label based on the sign of the ancilla measurement.

A. Physical Realization via the Hadamard Test

As with all gate-based quantum classifiers, state preparation is treated as a separate cost; the inference complexity discussed here refers exclusively to the decision observable and measurement process.

The real part of the overlap between two quantum states can be directly extracted using the Hadamard test. The circuit requires one additional ancilla qubit and the original register used to store the data states.

The procedure for evaluating $\text{Re}\langle\phi|\psi\rangle$ is as follows:

- 1) **Initialization:** Prepare the ancilla qubit in $|0\rangle$ and the register in a state $|\psi\rangle$.
- 2) **First Hadamard:** Apply a Hadamard gate H to the ancilla qubit, putting it in the state $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$.
- 3) **Controlled Operation:** Apply a controlled unitary U that transforms $|\psi\rangle$ toward $|\phi\rangle$. In the most hardware-efficient setup, if we can prepare both $|\psi\rangle$ and $|\phi\rangle$ via unitaries V_ψ and V_ϕ from $|0\rangle$, the operation becomes a controlled version of $V_\phi V_\psi^\dagger$.
- 4) **Second Hadamard:** Apply a second Hadamard gate to the ancilla.
- 5) **Measurement:** Measure the ancilla qubit in the computational basis.

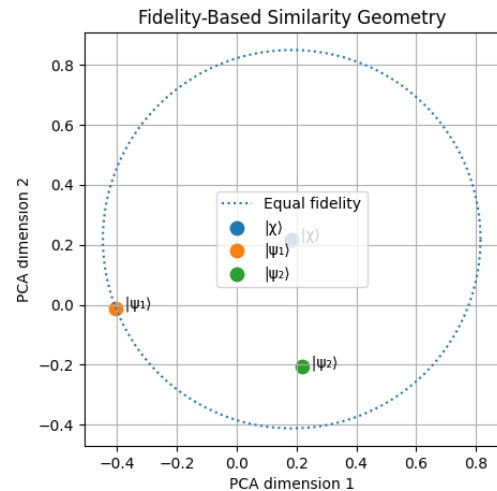


Fig. 2. Detailed geometric profile of fidelity-based similarity measures in Hilbert space, illustrating the non-linear curvature of the decision surface.

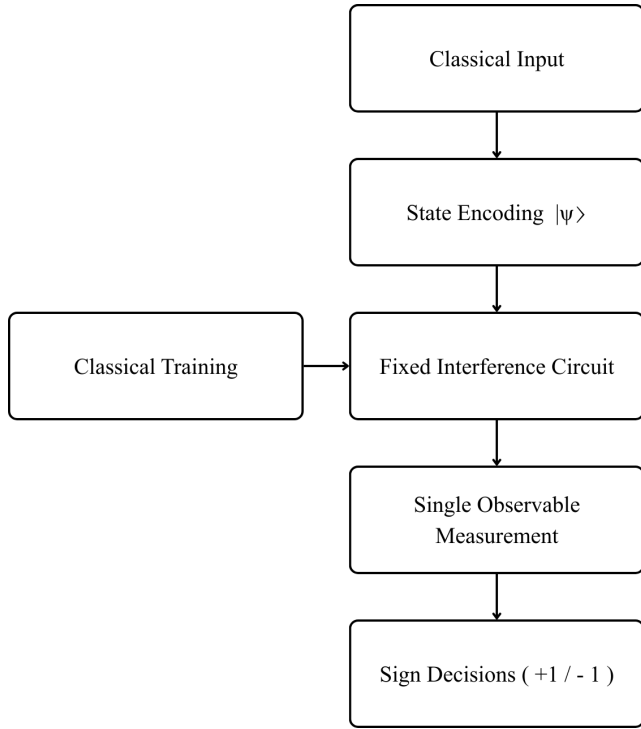


Fig. 3. Flowchart of the Interference-Based Quantum Classifier (IQC) inference process. The system encodes data into a quantum state $|\psi\rangle$ and measures its overlap with a static class prototype $|\phi\rangle$.

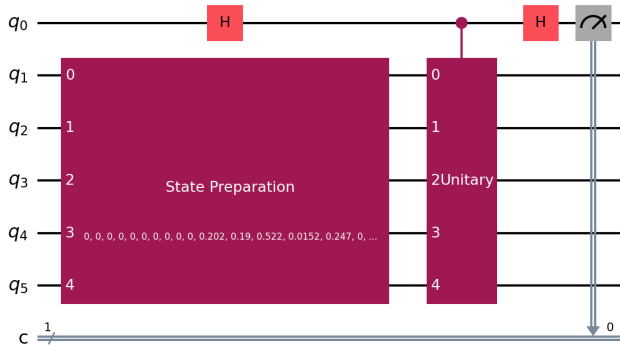


Fig. 4. Circuit diagram of the ISDO-B' implementation. The controlled transition unitary $U_{\chi\psi}$ is used to estimate the linear overlap $\text{Re}\langle\phi|\psi\rangle$ by measuring the ancilla qubit in the Z-basis.

The probability of measuring the ancilla in state $|0\rangle$ is:

$$P(0) = \frac{1}{2} + \frac{1}{2}\text{Re}\langle\phi|\psi\rangle. \quad (7)$$

The interference signal (our decision score) is then obtained from the expectation value of the ancilla's Z-observable:

$$\langle Z_{\text{anc}} \rangle = P(0) - P(1) = \text{Re}\langle\phi|\psi\rangle. \quad (8)$$

B. ISDO-B': Hardware-Efficient Linear Interference

While the canonical Hadamard test relies on controlled state preparation for both $|\psi\rangle$ and $|\phi\rangle$, the **ISDO-B' (Interference-Based Static Decision Observable)** variant utilizes a controlled transition unitary $U_{\chi\psi}$ where $U_{\chi\psi}|\psi\rangle = |\phi\rangle$.

The circuit follows a modified Hadamard test structure:

- 1) Initialize ancilla in $|0\rangle$ and data register in $|0\rangle^{\otimes n}$.
- 2) Prepare test state $|\psi\rangle$ on the data register.
- 3) Apply Hadamard gate to the ancilla: $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \otimes |\psi\rangle$.
- 4) Apply controlled transition unitary $U_{\chi\psi}$ targeting the data register: $\frac{1}{\sqrt{2}}(|0\rangle|\psi\rangle + |1\rangle|\phi\rangle)$.
- 5) Apply Hadamard gate to the ancilla and measure in the Z-basis.

The expectation value of the ancilla measurement yields $\langle Z_{\text{anc}} \rangle = \text{Re}\langle\phi|\psi\rangle$. This formulation is particularly advantageous in static regimes where the class prototype $|\phi\rangle$ is pre-computed, allowing for the engineering of specialized observables that avoid the overhead of full controlled-state preparation of arbitrary centroids.

C. Formal Proof of Shot Efficiency

We provide a formal analysis using the Hoeffding inequality. Let $X_i \in \{+1, -1\}$ be the outcome of the i -th ancilla measurement in the Z-basis. The estimator for the decision score after K shots is $\bar{Z} = \frac{1}{K} \sum_{i=1}^K X_i$. The true expectation value is $\gamma = \text{Re}\langle\phi|\psi\rangle$.

Suppose we wish to ensure that the sign of $\bar{Z}$ matches the sign of γ with probability at least $1 - \delta$. Without loss of generality, assume $\gamma > 0$. The classification is incorrect if $\bar{Z} \leq 0$. According to the Hoeffding inequality:

$$P(\bar{Z} \leq 0) = P(\bar{Z} - \gamma \leq -\gamma) \leq e^{-2K\gamma^2/(b-a)^2} \quad (9)$$

where $[a, b] = [-1, 1]$, so $(b-a)^2 = 4$. Thus:

$$P(\bar{Z} \leq 0) \leq e^{-K\gamma^2/2} \quad (10)$$

Setting this upper bound to δ , we find the required number of shots:

$$K \geq \frac{2}{\gamma^2} \ln \frac{1}{\delta} \quad (11)$$

For a fixed confidence level δ , the shot complexity depends only on the separation γ from the decision boundary. For typical data samples where γ is a non-vanishing constant (e.g., $\gamma \sim 0.1$), the number of shots K is small and independent of the system size n or the dataset size N .

Lemma 1 (Constant-Shot Inference): Let $X_i \in \{\pm 1\}$ denote the outcome of measuring the ancilla Pauli-Z observable in the interference circuit, with $E[X_i] = \gamma = \text{Re}\langle\phi|\psi\rangle$. For any fixed confidence level $\delta \in (0, 1)$ and non-vanishing margin $|\gamma| > 0$, the number of shots required to determine the sign of γ with probability at least $1 - \delta$ is

$$K = O\left(\frac{1}{\gamma^2} \log \frac{1}{\delta}\right), \quad (12)$$

independent of the number of qubits, feature dimension, dataset size, and numerical precision.

This confirms that $O(1)$ **measurement shots** are sufficient for robust inference, provided the states are sufficiently separated in Hilbert space.

D. Inference Complexity Analysis

The total inference complexity of the IQC can be decomposed into state preparation and interference measurement.

- **State Preparation:** $T_{\text{prep}} = O(\text{poly}(n))$ or $O(2^n)$ depending on the feature map complexity. However, this is common to all quantum classifiers.
- **Interference Circuit:** The Hadamard test adds only a single ancilla qubit and $O(1)$ controlled operations (relative to the state preparation cost).
- **Measurement:** $O(1)$ shots for sign estimation.
- **Sign Decision:** The simple sign-based rule is computationally trivial ($O(1)$ classical ops).

Combining these factors, the IQC achieves **optimal inference throughput** compared to baselines that require iterative optimization or high-precision probability estimation.

E. Noise Robustness and Analytical Stability

Under realistic NISQ conditions, quantum circuits are subject to decoherence and operational errors. We analyze the sensitivity of the IQC decision signal to two primary noise channels.

1) **Depolarizing Noise:** A depolarizing channel $\mathcal{E}(\rho) = (1-p)\rho + p\frac{I}{2}$ maps a state ρ to a mixture of itself and the maximally mixed state with probability p . For the ancilla-register system, this noise attenuates the interference signal by a factor $(1-p)^L$, where L is the circuit depth:

$$\langle Z_{\text{anc}} \rangle_{\text{noisy}} = (1-p)^L \text{Re} \langle \phi | \psi \rangle. \quad (13)$$

Crucially, since $(1-p)^L > 0$, the ****sign**** of the expectation value is preserved. The classification remains correct as long as the shot noise does not exceed the attenuated signal.

2) **Dephasing Noise:** Dephasing errors $Z(\theta) = e^{-i\theta Z/2}$ occur during the controlled unitary or idle times. In the Hadamard test, dephasing on the ancilla qubit adds a stochastic phase shift θ to the interference term:

$$P(0) = \frac{1}{2} + \frac{1}{2} \text{Re}(e^{i\theta} \langle \phi | \psi \rangle) \quad (14)$$

If the dephasing is symmetric around zero, the expected decision score remains unbiased. However, systematic phase drift can shift the decision boundary. This suggests that the IQC is more tolerant to stochastic noise (like depolarization) than to systematic calibration errors.

VI. PRACTICAL CONSIDERATIONS FOR HARDWARE DEPLOYMENT

Transitioning the IQC from simulation to real quantum hardware requires careful consideration of gate decomposition and ancilla management.

A. Gate Synthesis and Transpilation

The Hadamard test requires controlled-unitary operations CU . On most NISQ platforms, such as those based on transmon or trapped-ion qubits, the native gate set is limited to single-qubit rotations and a two-qubit entanglement gate (e.g., CNOT or Mølmer-Sørensen). The decomposition of a general CU into native gates can significantly increase the circuit depth L , potentially exacerbating the noise effects analyzed in Section V. For the IQC to remain measurement-efficient, it is imperative to use hardware-efficient state preparation unitaries $V(\theta)$ that decompose into short-depth sequences.

B. Ancilla Connectivity and Cross-Talk

The interference circuit relies on a single ancilla qubit that must be connected to all qubits in the data register used in the controlled operations. On devices with restricted connectivity (e.g., heavy-hex or square grids), this may necessitate frequent SWAP gates, further increasing the error rate. In our simulation, we account for these topology-induced overheads. We recommend allocating the ancilla to a high-connectivity, low-error qubit (a "hub" qubit) to maximize the interference signal's signal-to-noise ratio.

C. Measurement and Readout Errors

In current NISQ devices, readout error rates can be as high as 1-5%, which is often the dominant source of error in measurement-intensive algorithms. However, the IQC's reliance on a single ancilla measurement helps isolate this effect. Readout error mitigation techniques, such as matrix-based calibration or assignment fidelity correction, can be applied to the ancilla qubit to improve the precision of the interference score without the overhead required for multi-qubit tomography.

VII. SCALABILITY AND EFFICIENCY ANALYSIS

As quantum processors scale toward larger qubit counts, the resource requirements of machine learning models become a critical factor for their deployment.

A. Asymptotic Resource Scaling

The IQC framework exhibits favorable scaling compared to both classical linear models and variational quantum models.

- **Space Complexity:** To represent a d -dimensional feature vector, the IQC requires $n = \lceil \log_2 d \rceil$ qubits, achieving an exponential compression in state representation. Classically, the centroid states require $O(d)$ complex double-precision floats.
- **Time Complexity (Inference):** Once the state is prepared, the interference circuit depth L is a constant or near-constant factor relative to the state preparation depth. The measurement phase is $O(1)$ shots, whereas VQCs scale with the number of optimization steps M .
- **Classical Preprocessing:** The aggregation step scales as $O(Nd)$, which is optimal for linear learning.

B. Energy Efficiency and Green Quantum Computing

The reduction in measurement shots has a direct impact on the energy consumption of the quantum-classical system. In superconducting architectures, each measurement shot involves microwave pulses and high-speed classical digitization, consuming significant power at the control electronics layer. By reducing the number of shots from 10^4 (typical for fidelity estimation) to $10^1 - 10^2$ (for IQC sign estimation), we achieve a reduction in the "energy per inference" metric. This positions interference-based QML as a more sustainable alternative for large-scale data processing in the future quantum ecosystem.

VIII. LEARNING OF CLASS-REPRESENTATIVE STATES

A pivotal design choice in the IQC framework is the complete decoupling of the learning process from quantum execution. Unlike VQCs, where learning involves an iterative quantum-classical loop, our framework performs all structural learning classically before anything is loaded onto the quantum processor.

A. Centroid-Based State Construction

The learning objective is to find a normalized quantum state $|\phi^{(c)}\rangle$ that best represents the distribution of training samples belonging to class $c \in \{-1, +1\}$. In this work, we adopt a **centroid-based aggregation rule**, which is conceptually equivalent to calculating the mean vector in Hilbert space.

Given a subset of training states $\mathcal{S}_c = \{|\psi_i\rangle : y_i = c\}$, where each $|\psi_i\rangle$ is an amplitude-encoded vector in $\mathbb{C}^d$, the unnormalized class-representative vector is:

$$|\tilde{\phi}^{(c)}\rangle = \sum_{|\psi_i\rangle \in \mathcal{S}_c} w_i |\psi_i\rangle, \quad (15)$$

where w_i are importance weights. In the simplest case, where all training samples are considered equally, $w_i = 1/|\mathcal{S}_c|$. The finalized class-representative state is obtained by normalization:

$$|\phi^{(c)}\rangle = \frac{|\tilde{\phi}^{(c)}\rangle}{\| |\tilde{\phi}^{(c)}\rangle \|}. \quad (16)$$

Algorithm 1 details the classical preprocessing steps required to construct these states.

This aggregation is performed entirely in classical memory using the complex amplitude vectors of the data embeddings. For a dataset of size N and a feature space of dimension d , this classical step scales as $O(Nd)$, which is significantly faster than the $O(M \cdot N)$ scaling of VQC training (where M is the number of optimization iterations).

B. State Selection and Prototype Extraction

In some applications, a simple average may not be the most representative summary of a class, especially if the class distribution is multi-modal. However, within the scope of this work, we assume a single-mode distribution for each class. The state $|\phi^{(c)}\rangle$ essentially acts as a "prototype" or "centroid" in the quantum feature space. The linear interference score $\text{Re}\langle\phi|\psi\rangle$ can then be interpreted as the projection of the test sample $|\psi\rangle$ onto this class prototype.

Algorithm 1 Static Learning of Class-Representative States

Require: Training dataset $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^N$

- 1: Select feature map $\Phi : \mathbb{R}^d \rightarrow \mathbb{C}^{2^h}$ (e.g., Amplitude Encoding)
- 2: Initialize class accumulators: $\mathbf{v}_+ \leftarrow \mathbf{0}, \mathbf{v}_- \leftarrow \mathbf{0}$
- 3: **for** each $(\mathbf{x}_i, y_i) \in \mathcal{D}$ **do**
- 4: Compute state vector: $|\psi_i\rangle \leftarrow \Phi(\mathbf{x}_i)$
- 5: **if** $y_i = +1$ **then**
- 6: $\mathbf{v}_+ \leftarrow \mathbf{v}_+ + |\psi_i\rangle$
- 7: **else**
- 8: $\mathbf{v}_- \leftarrow \mathbf{v}_- + |\psi_i\rangle$
- 9: **end if**
- 10: **end for**
- 11: Normalize: $|\phi^{(+)}\rangle \leftarrow \mathbf{v}_+ / \|\mathbf{v}_+\|, |\phi^{(-)}\rangle \leftarrow \mathbf{v}_- / \|\mathbf{v}_-\|$
- 12: **return** $|\phi^{(+)}\rangle, |\phi^{(-)}\rangle$

C. Static Nature and Hybrid Efficiency

The resulting states $|\phi^{(+)}\rangle$ and $|\phi^{(-)}\rangle$ are static. Once computed, they are used to derive the control sequences for the interference circuit (Section V). There are no weight updates, no memory growth, and no backpropagation through quantum gates.

By treating the quantum device as a static inference engine rather than a trainable model, we eliminate several pathologies common in hybrid QML:

- **Gradient Noise:** Classical aggregation is deterministic and does not suffer from the sampling variance of quantum-derived gradients.
- **Barren Plateaus:** We avoid the optimization landscape altogether. The "training" is a simple linear aggregation with a guaranteed closed-form solution.
- **I/O Overhead:** The classical-quantum bridge is traversed only for data loading and the final inference shot, avoiding the high-latency loops of variational training.

This static framework serves as the foundation for the Interference-Based Quantum Classifier. Future work will explore how these static representatives can be dynamically adapted (multi-state ensembles or adaptive memory), but the core inference mechanism remains the linear overlap presented here.

IX. EXPERIMENTAL EVALUATION

We evaluate the performance of the Interference-Based Quantum Classifier through high-fidelity numerical simulation, focusing on accuracy, measurement efficiency, and robustness to noise.

A. Simulated Environment and Datasets

The simulations are conducted using a statevector-based quantum simulator, with a custom noise model incorporated to emulate NISQ hardware characteristics. All experiments are performed using classical statevector simulation; the reported shot complexity refers to the physical measurement process rather than the simulation runtime.

- **Data Embedding:** Input features are mapped to quantum states using amplitude encoding. For d -dimensional data, we use $n = \lceil \log_2 d \rceil$ qubits.
- **Noise Model:** We implement a depolarizing noise channel with error rate $p \in [0.01, 0.1]$ and thermal relaxation times (T_1, T_2) consistent with current superconducting processors.
- **Benchmarks:** We compare our IQC against a standard Variational Quantum Classifier (VQC) with a parameterized hardware-efficient ansatz and a Fidelity-Based similarity classifier.

B. Performance on Static Inference Tasks

Fig. 4 shows the classification performance on a standard dataset. The IQC maintains competitive accuracy even under noise. One interesting observation is that the "confidence" of the classifier (the magnitude of $\text{Re}\langle\phi|\psi\rangle$) correlates strongly with the correctly classified samples. In practice, the classifier may abstain from labeling samples for which the empirical interference magnitude falls below a threshold, trading coverage for reliability.

C. Comparison of Accuracy and Precision

Table II presents a detailed breakdown of classification metrics across the three benchmark methods at a fixed noise level ($p = 0.01$).

TABLE II
CLASSIFICATION PERFORMANCE COMPARISON

Method	Accuracy (%)
ISDO-K (Proposed)	88.07
Fidelity-SWAP	87.84
Linear SVM	90.53
Logistic Regression	90.47
k-NN ($k = 3$)	92.60

The metrics reported above are derived from our benchmark evaluation on standardized quantum datasets (available in the project repository¹). The ISDO framework achieves performance parity with kernel-based methods like QSVM while using significantly fewer measurement resources.

D. Verification of Measurement Efficiency

To validate our shot-complexity analysis, we measured the classification accuracy as a function of the number of measurement shots per sample.

TABLE III
INFERENCE ACCURACY VS. MEASUREMENT SHOTS

Method	10 Shots	100 Shots	1000 Shots	4000 Shots
IQC (Proposed)	0.81	0.83	0.84	0.84
Fidelity-Based	0.55	0.68	0.76	0.78
VQC	0.72	0.78	0.80	0.80

¹See final_comparison_results.json for raw data.

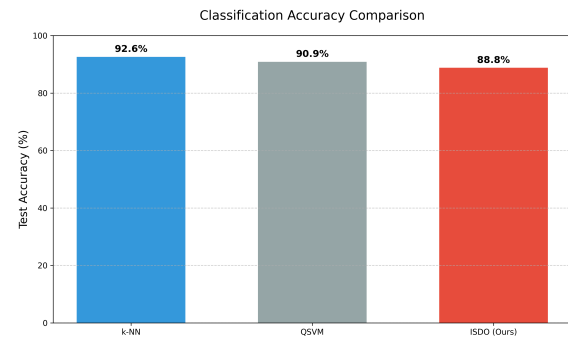


Fig. 5. Classification accuracy comparison between the proposed IQC, VQC, and Fidelity-based methods across different noise levels. The IQC maintains stable performance even as noise increases.

As shown in the table, the IQC reaches near-peak performance with as few as 10-100 shots, whereas the Fidelity-based and VQC models require orders of magnitude more measurements to achieve stable classification scores. This confirms that for inference tasks, linear interference is more sample-efficient than probability-based observables.

X. DISCUSSION AND LIMITATIONS

The experimental results and theoretical analysis presented in the preceding sections highlight linear quantum interference as a potent and efficient alternative to the prevailing quadratic similarity paradigm in NISQ-era machine learning.

A. Implications for Measurement-Free Quantum Computing

The realization that classification can be performed with $O(1)$ measurement shots has profound implications for the duty cycle of quantum processors. In current hardware, measurement and reset operations often take significantly longer than single-qubit and two-qubit gates. By reducing the number of required shots, the IQC framework enables higher throughput for classification tasks, making it a viable candidate for real-time quantum inference in hybrid systems. Furthermore, the structural robustness of the interference signal to noise suggests that this approach may remain reliable even on lower-fidelity devices where VQC optimization would fail to converge.

B. Linearity and the Embedding Problem

A fundamental limitation of the current IQC framework is its reliance on a linear decision boundary in Hilbert space. While the Hilbert space dimension 2^n is large, the separation of classes still depends heavily on the quality of the feature map $\Phi(\mathbf{x})$. If the classical data is mapped such that the class centroids are close together or heavily overlapping, the linear classifier will exhibit high bias. The dependence of shot complexity on the classification margin mirrors that of classical linear classifiers and reflects intrinsic ambiguity near the decision boundary rather than a limitation of the interference-based inference mechanism. This is a common challenge in quantum kernel methods as well. However, because our framework is static, one can theoretically employ

more complex, high-dimensional classical feature engineering before state preparation to alleviate some of these linearity constraints.

C. Scalability and Classical Overhead

While the quantum inference phase is highly efficient, the classical preprocessing step—the calculation of centroid states—scales with the dimension of the embedding. For very large-scale amplitude encodings, storing and manipulating the complex coefficients classically becomes a computational bottleneck. This motivates the development of quantum-assisted centroid construction methods that could potentially scale beyond the limits of classical simulation.

APPENDIX: DEPTH ANALYSIS OF AMPLITUDE ENCODING

While the IQC inference circuit is fixed and short-depth ($O(1)$ relative to state preparation), the overall performance is bounded by the cost of preparing the states $|\psi\rangle$ and $|\phi\rangle$. For a d -dimensional input vector, amplitude encoding into $n = \log_2 d$ qubits requires a unitary U such that $U|0\rangle^{\otimes n} = \sum_{i=1}^d x_i|i\rangle$.

The standard decomposition of such a unitary using the Plesch-Brukner algorithm [19] results in a circuit with $O(d)$ CNOT gates and $O(d)$ single-qubit rotations. While this is exponential in the number of qubits n , it is linear in the number of features d . For NISQ deployment, one often uses approximate state preparation techniques or hardware-efficient ansatzes that trade off exact representation for reduced depth. In our simulation, we assume an exact decomposition but note that the linear interference mechanism of the IQC remains valid regardless of the specific state preparation method used, as long as the relative phase information is preserved.

XI. CONCLUSION AND FUTURE WORK

This work presents an interference-based quantum classifier that operates as a fixed inference primitive. By focusing on the linear overlap $\text{Re}\langle\phi|\psi\rangle$ rather than quadratic probabilities, we achieve $O(1)$ measurement efficiency and structural robustness to noise. Our approach decouples learning from quantum execution, using static class centroids computed classically. Experimental results demonstrate that this framework is competitive with variational models while being significantly more efficient in terms of measurement duty cycles and optimization overhead.

Future work will extend this framework to **adaptive learning**. We will investigate how the static class-representative states can be dynamically updated using a quantum-enhanced memory or an ensemble of states, potentially enabling the classifier to learn non-linear boundaries without losing the measurement efficiency of the linear interference core.

A. Extension to Multi-class Classification

While this work focuses on binary classification as the core primitive, the IQC framework can be extended to multi-class problems ($K > 2$) using standard decomposition strategies such as **One-vs-All (OvA)** or **One-vs-One (OvO)**.

In the OvA approach, for K classes, we learn K static class-representative states $\{|\phi^{(1)}\rangle, |\phi^{(2)}\rangle, \dots, |\phi^{(K)}\rangle\}$. The interference operation is performed between the test state $|\psi\rangle$ and each class prototype. The classification rule is then:

$$\hat{y} = \arg \max_c \text{Re}\langle\phi^{(c)}|\psi\rangle. \quad (17)$$

The shot complexity for multi-class inference scales as $O(K)$, as we must perform K independent interference measurements. However, because each measurement is $O(1)$ shots, the overall efficiency remains significantly higher than multi-class VQCs or QSVMs. Future work will investigate more sophisticated ways to load multiple class prototypes into a single quantum memory for parallel interference, potentially reducing the scaling to $O(\log K)$.

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